

01_validate_and_clean.R

Complete source code for reference. The original .R file must remain in GitHub for execution.

```
# Validate, clean and prepare the advertising campaign data
# Base R only: no external packages are required.

data <- read.csv("data/advertising_campaigns.csv", stringsAsFactors = FALSE)
required <- c("date", "platform", "campaign_type", "audience", "region", "spend", "impressions", "clicks", "conversions", "revenue")
stopifnot(all(required %in% names(data)))

data$date <- as.Date(data$date)
stopifnot(all(!is.na(data$date)))
numeric_cols <- c("spend", "impressions", "clicks", "conversions", "revenue")
for (v in numeric_cols) stopifnot(all(is.finite(data[[v]])))
stopifnot(all(data$spend >= 0), all(data$impressions > 0), all(data$clicks >= 0),
           all(data$conversions >= 0), all(data$revenue >= 0), all(data$clicks <= data$impressions))

data$ctr <- data$clicks / data$impressions
data$conversion_rate <- ifelse(data$clicks > 0, data$conversions / data$clicks, 0)
data$cpc <- ifelse(data$clicks > 0, data$spend / data$clicks, NA_real_)
data$roas <- ifelse(data$spend > 0, data$revenue / data$spend, NA_real_)

# Fixed, reproducible 80/20 split used by the predictive model.
set.seed(42)
idx <- sample(seq_len(nrow(data)), floor(0.80 * nrow(data)))
data$split <- ifelse(seq_len(nrow(data)) %in% idx, "train", "test")

dir.create("outputs", showWarnings = FALSE)
dir.create("outputs/plots", showWarnings = FALSE, recursive = TRUE)
write.csv(data, "outputs/clean_advertising_campaigns.csv", row.names = FALSE)
cat("Validation and cleaning completed:", nrow(data), "rows\n")
cat("Training rows:", sum(data$split == "train"), "Test rows:", sum(data$split == "test"), "\n")
```